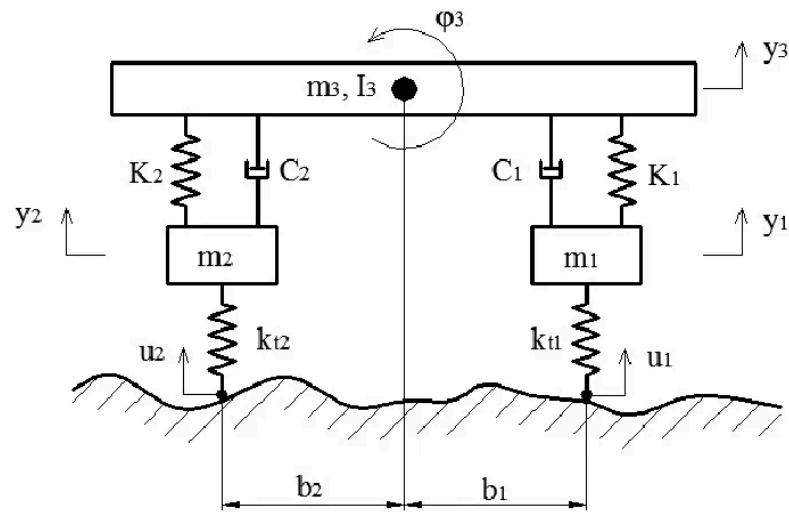


VEHICLE DYNAMICS OPTIMIZATION OF A HALF CAR MODEL

USING BAYESIAN OPTIMISATION



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1. INTRODUCTION

Vehicle dynamics is part of automotive design that dictates the behaviour of a vehicle under various performance conditions. It has a heavy influence over many different characteristics of a car, like comfort, handling, traction, braking etc. Since so many different characteristics are dependent upon the choices made during suspension design, it is one of the most crucial and delicate sub-systems and requires engineers to make informed decisions.

Suspension design is a fundamentally multi-objective engineering problem. While a softly sprung setup will absorb variations in the road profile and provide better superior ride comfort, it causes higher tyre dynamic load variations, leading to poor handling characteristics (tyre grip). Conversely a stiffly sprung setup will minimise tyre load variations, thus ensuring consistent and stable tyre grip. A major drawback of this setup is that it lacks the ability to absorb the vibrations caused by undulations in the road surface, thus drastically reducing passenger comfort.

This report presents a framework that leverages Bayesian Optimization for the simultaneous optimization of five influential variables on a half car model. The optimiser minimises a composite loss function which combines three normalised sub-objectives.

2. THE MODEL

2.1. Degrees of Freedom

The half car model can be imagined as a longitudinal cross section of the vehicle. It captures the four most important degrees of freedom for ride and handling analysis. The degrees of freedom in this model are:

- Sprung Mass Heave z_s – vertical translation of the vehicle body at the centre of gravity of the model.
- Sprung Mass Pitch θ – rotation of the vehicle body about a transverse axis passing through the centre of gravity.
- Front Wheel Hub Heave z_f – vertical displacement of the front unsprung mass.
- Rear Wheel Hub Heave z_r – vertical displacement of the rear unsprung mass.

2.2. Equations of Motion

The equations of motion are derived by applying *Newton's Second Law of Motion* to

each body. All the displacements are measured with respect to static equilibrium, this helps eliminate the gravity component, thus simplifying the equations. The sign convention employed for the simulation considers a nose-down condition (dive) to be positive pitch and upwards displacement of a body to be positive heave.

For the sprung body:

$$m.z_s = -F_{\{s,f\}} - F_{\{s,r\}}$$

$$I_y.\dot{\theta} = a.F_{\{s,f\}} - b.F_{\{s,r\}}$$

For the unsprung mass:

$$m_f.z_f = F_{\{s,f\}} - F_{\{t,f\}}$$

$$m_r.z_r = F_{\{s,r\}} - F_{\{t,r\}}$$

The suspension force at each corner is modelled as a *Kelving-Voigt spring damper element*:

$$F_{\{s,f\}} = k_f.d_f + c_f.\dot{d}_f$$

The suspension deflections from equilibrium and their respective rates are:

$$d_f = z_s - a.\theta - z_f$$

$$\dot{d}_f = \dot{z}_s - a.\dot{\theta} - \dot{z}_f$$

The tyre is modelled as a pure linear spring that connects the wheel hub to the road without any damping:

$$F_{\{t,f\}} = k_{\{t,f\}}.(z_f - z_{\{r,f\}})$$

2.3. Fixed Parameters

Parameter	Symbol	Value	Basis
Sprung mass	m	1200 kg	Midsized saloon
Wheelbase	L	2.70 m	Midsized platform
Pitch moment of inertia	I_y	1800 kg·m ²	$m \cdot (0.387L)^2$ approx.
Front unsprung mass	m_f	40 kg	Wheel + hub + brake
Rear unsprung mass	m_r	35 kg	Wheel + hub + brake

Parameter	Symbol	Value	Basis
Front tyre stiffness	$k_{\{t,f\}}$	200 kN/m	Standard radial tyre
Rear tyre stiffness	$k_{\{t,r\}}$	200 kN/m	Standard radial tyre
Vehicle speed	v	20 m/s	72 km/h cruise

Table 1: Fixed vehicle parameters.

3. ROAD PROFILE

This simulation employs a deterministic road profile model with a well defined geometry. This ensures repeatability of the simulation, allowing the engineer to analyse and compare results for different inputs.

Each feature is defined in the spatial domain and evaluated at each time step via the front-axle position $\mathbf{x} = \mathbf{v} \cdot \mathbf{t}$. The rear axle receives the same profile with a pure integer-sample time delay $\Delta t = L/v$.

Feature	Shape	Amplitude	Length	Mathematical Form
Speed bump	Half-cosine hump	80 mm	500 mm	$z = A/2 \cdot (1 - \cos(2\pi\xi/L))$
Pothole	Inverted half-cosine	50 mm	300 mm	$z = -A/2 \cdot (1 - \cos(2\pi\xi/L))$

Table 2: Deterministic road feature library.

$$z = -A/2 \cdot (1 - \cos(2\pi\xi/L))$$

Both profiles use the half-cosine shape, which has zero displacement and zero slope at both ends of the feature window. This ensures continuity of the road profile and its first derivative at the feature edges, which is much closer to reality than having a step bump in the road profile. The feature is placed at 50 m from the origin of the simulation. This means that the front axle will encounter the obstacle at about 2.5 s from the start. Considering the time taken for the rear axle to encounter the obstacle, about 12.5 s of time still remains in the simulation time frame for free decay.

4. LOSS FUNCTION BAYESIAN

4.1. Three Layer Architecture

The loss function 'J' is responsible for mapping the proposed parameter set of {k_f, k_r, c_f, c_r, a} to a single scalar that Bayesian optimization seeks to minimize. It is constructed in three layers to ensure that the mode weights have a physically meaningful and dimensionally consistent interpretation.

Layer 1:

This is the raw metrics layer. Three fundamental performance metrics are extracted from each ODE simulation:

1. Comfort – RMS of heave acceleration.

$$\text{rms}(\ddot{z}_s) \text{ [m/s}^2\text{]}$$

2. Handling – Sum of RMS tyre dynamic forces at both axles

$$\text{rms}(F_{\{t,f\}}) + \text{rms}(F_{\{t,r\}}) \text{ [N]}$$

3. Pitch – RMS pitch angular acceleration scaled to equivalent linear acceleration.

$$\text{rms}(\theta) \cdot (L/2) \text{ [m/s}^2\text{]}$$

Layer 2:

This is the normalisation layer. Each raw metric is divided by a reference value in the same physical units, producing dimensionless sub-objectives:

$$J_c = \text{rms}(\ddot{z}_s) / \text{ref}_c$$

$$J_h = [\text{rms}(F_{\{t,f\}}) + \text{rms}(F_{\{t,r\}})] / \text{ref}_h$$

$$J_p = \text{rms}(\theta) \cdot (L/2) / \text{ref}_p$$

This is an important layer because it prevents any one parameter from numerically dominating the loss. This step gives relevance to the assigned weights.

Reference	Value	Method	Justification
ref_c (comfort)	1.50 m/s ²	ISO standard	ISO 2631-1 'uncomfortable' onset boundary
ref_h (handling)	Computed	Baseline simulation	Baseline params score exactly 1.0; no arbitrary constant

Reference	Value	Method	Justification
ref_p (pitch)	Computed	Baseline simulation	Baseline params score exactly 1.0; consistent with ref_h approach

Table 3 — Normalisation reference values and their sources

Layer 3:

This is the weighted sum layer. The composite loss is a combination of the three normalised sub-objectives.

$$J(p) = w_c \cdot J_c + w_h \cdot J_h + w_p \cdot J_p$$

Since all the weights are non-negative and their summation is one, J is bounded by zero (physically perfect suspension).

4.2. Drive Mode Weights

Mode	w_c (Comfort)	w_h (Handling)	w_p (Pitch)
Luxury	0.70	0.20	0.10
Sport	0.20	0.55	0.25

Table 4 — Mode weight sets. Both rows sum to 1.0.

In Luxury mode the iso-loss hyperplane is steeply inclined toward the comfort axis, and the optimiser finds soft, well-isolated tunings. In Sport mode the hyperplane tilts toward the handling axis, rewarding suspensions that minimize dynamic tyre-load variation at the expense of body acceleration.

5. BAYESIAN OPTIMISATION FRAMEWORK

5.1. Design Variables and Search Bounds

Five different variables with physically defined bounds are passed to the bayesopt function:

Variable	Symbol	Lower	Upper	Transform	Physical Basis
Front spring	k_f	8 kN/m	60 kN/m	Logarithmic	f_n = 1.0–2.25 Hz
Rear spring	k_r	8 kN/m	60 kN/m	Logarithmic	f_n = 1.0–2.25 Hz
Front damper	c_f	500 N·s/m	5000 N·s/m	Logarithmic	ζ = 0.2–0.5
Rear damper	c_r	500 N·s/m	5000 N·s/m	Logarithmic	ζ = 0.2–0.5
CG position	a	0.30·L	0.70·L	Linear	Axle load ≥30%

Table 5 — Optimisation variables with bounds and transforms.

The logarithmic transform on the stiffness and damping variables is crucial for optimizer performance. These parameters span approximately one order of magnitude, and the loss function surface is far smoother in log-space than linear space. The Gaussian process surrogate builds a more accurate model of the objective landscape with the same number of evaluations when the variables are log-transformed, resulting in better convergence within the 80-evaluation budget.

5.2. Algorithm Configuration

Setting	Value and Rationale
Acquisition function	Expected Improvement Plus (EI+) balances exploration and exploitation; '+' jitter prevents premature clustering
IsObjectiveDeterministic	true because ODE simulation is noise-free; GP interpolates exactly through observations
MaxObjectiveEvaluations	80 exceeds empirical; 10–20 evaluations per variable rule (5 variables → min 50)
NumSeedPoints	10 Latin hypercube samples bootstrap the GP before informed sampling begins

Table 6 — Bayesian optimization configuration.

5.3. Algorithm Loop

The optimization proceeds in two phases. In the seed phase (evaluations 1–10), ten quasi-random Latin hypercube points are evaluated to provide initial coverage of the five-dimensional search space and bootstrap the Gaussian process hyperparameter estimation.

In the iterative phase (evaluations 11–80), the loop repeats: fit the GP surrogate to all observations, maximise EI+ over the surrogate to identify the next candidate p^* , evaluate $\text{loss_halfcar}(p^*)$ by running the full ODE simulation, and adding the result to the observation set. The surrogate progressively sharpens, uncertainty estimates become honest, and the minimum observed loss converges.

6. MATLAB IMPLEMENTATION

In order to avoid cross-file dependency issues, the implementation is structured as a single MATLAB script with local functions.

- **bayesopt** calls **loss_halfcar** once per evaluation i.e. a total of 80 times.
- **loss_halfcar** calls **simulate_halfcar** to run the ODE and extract RMS metrics, then applies normalisation and weighting.
- **simulate_halfcar** calls **ode45** with **halfcar_ode** as the right-hand side function (called ~50,000–100,000 times per simulation).

7. RESULTS AND DISCUSSION

7.1. Baseline Normalisation References

Before the optimization begins, a baseline simulation is run with all five variables at the geometric midpoint of their search bounds. This produces the reference values used to normalise the handling and pitch sub-objectives:

- $\text{ref_c} = 1.50 \text{ m/s}^2$ (fixed ISO 2631-1 threshold)
- ref_h = computed from baseline simulation [N]
- ref_p = computed from baseline simulation [m/s^2]

By construction the baseline suspension scores exactly 1.0 on the handling and pitch sub-objectives, making the mode weights directly interpretable.

7.2. Optimal Parameter Discussion

The optimal parameters produced by the Bayesian optimization for each mode reflect the physical consequences of the weight vectors in a consistent and interpretable way.

When Luxury mode is selected, the optimizer is driven to minimize the body heave acceleration. This is achieved by reducing the spring stiffness of the suspension, thus reducing the body bounce natural frequency towards the lower ranges where road energy is less efficiently transmitted to the spring mass of the vehicle. Elevated damping ratios also assist by suppressing resonant amplification after the obstacle has been encountered by the respective axle. The CG position tends toward a value that balances front and rear suspension travel, preventing one end from bottoming out while the other is lightly loaded.

In Sport mode, the optimizer prioritises minimising dynamic tyre load variation. This demands higher spring stiffnesses which keeps wheel displacement relative to the body small so the tyre follows the road surface more closely. A CG position that balances the diving and squatting behaviour of the car such that neither axle is disproportionately loaded during the transient event.

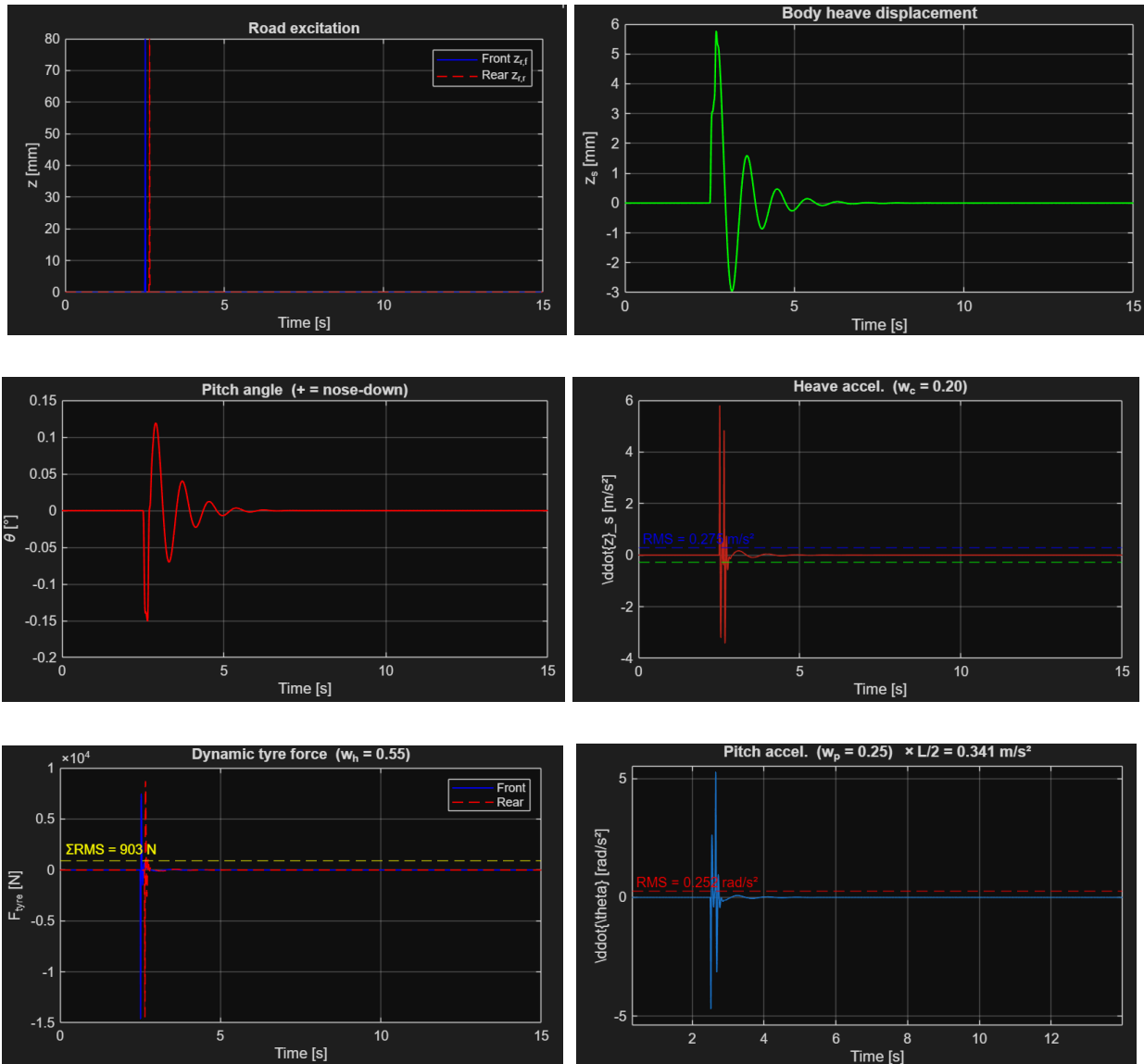
7.3. Convergence

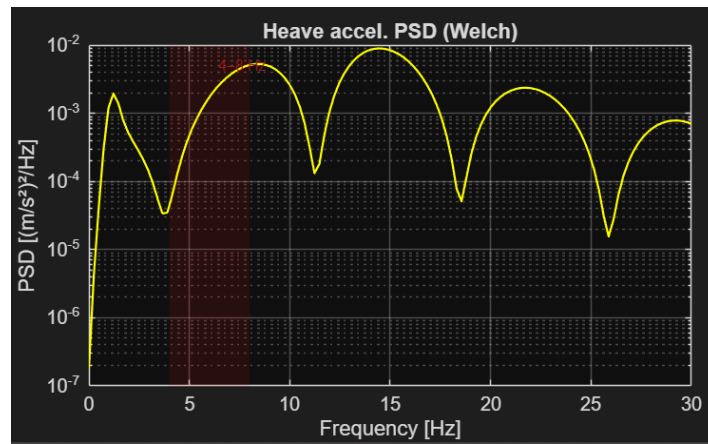
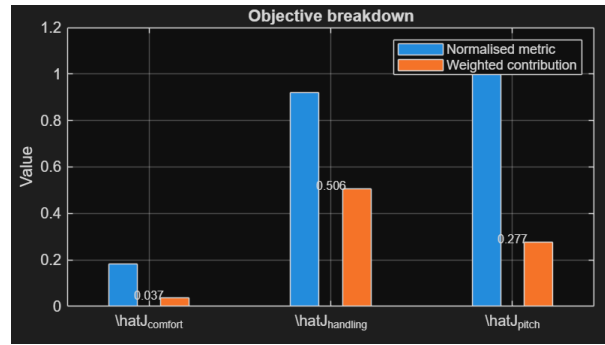
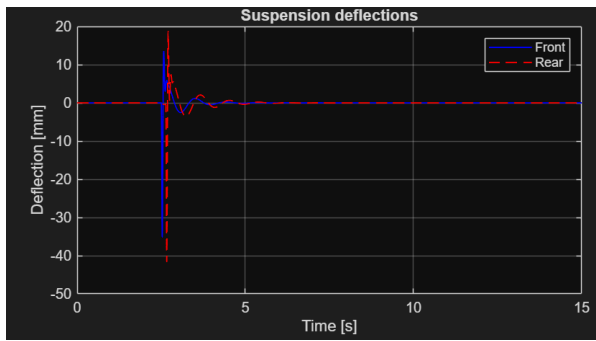
The minimum observed objective plotted against the number of function evaluations should show a steep descent during the seed phase (evaluations 1–10) followed by progressively smaller improvements as the optimiser homes in on the minimum. A curve that is still decreasing significantly at evaluation 80 indicates an insufficient evaluation budget; a flat curve from evaluation 30 onward indicates good convergence within the budget.

The EI+ acquisition function prevents premature convergence by occasionally sampling away from the current best region when the surrogate uncertainty warrants it. This behaviour appears as occasional upward steps in the objective history plot during the iterative phase before the minimum is recovered.

8. PLOTS

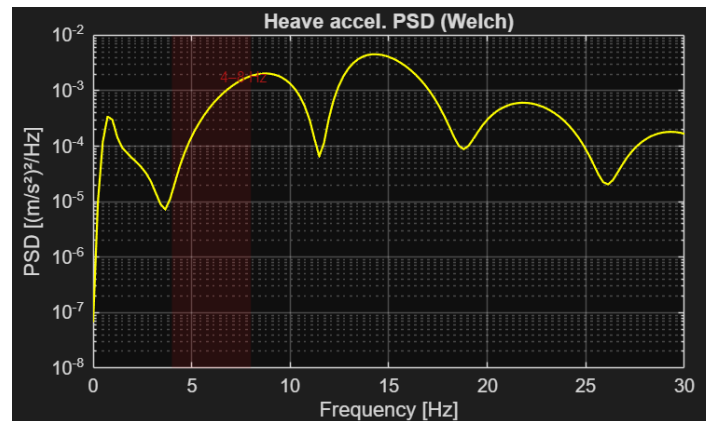
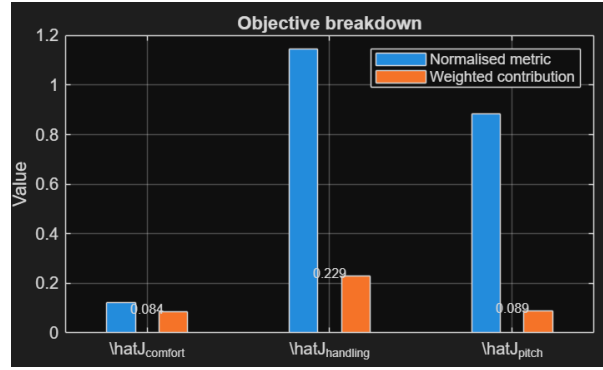
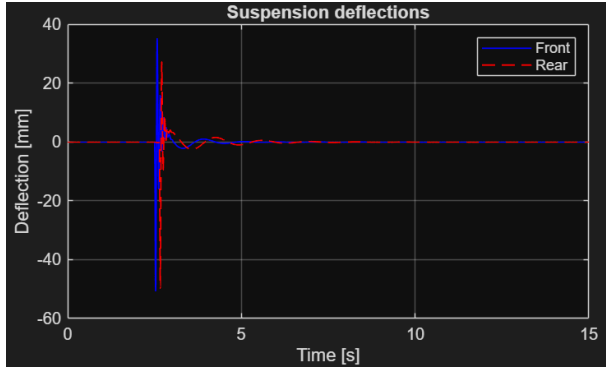
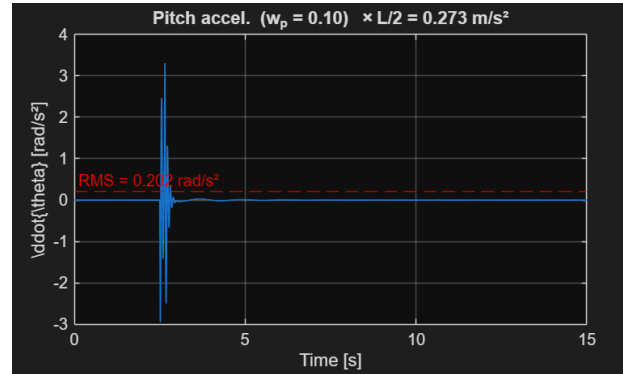
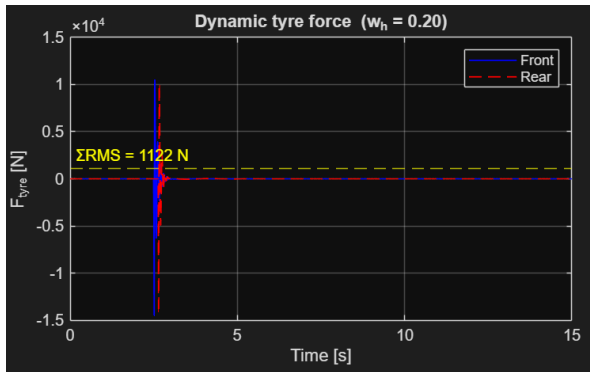
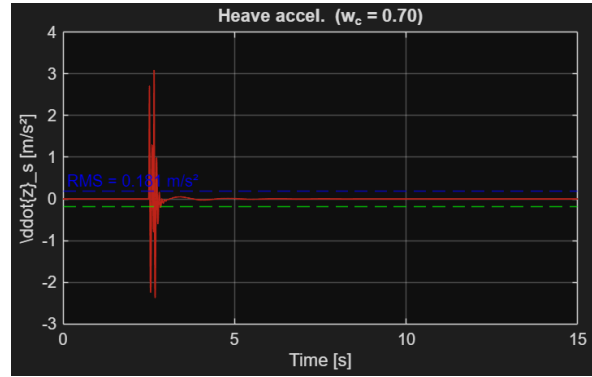
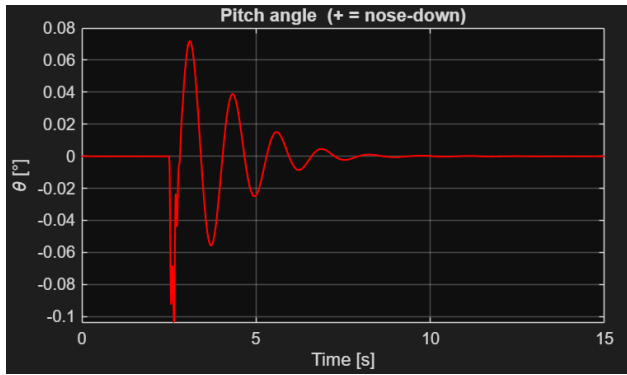
Sport Mode:





Luxury Mode:





9. CONCLUSION

This project demonstrated a complete and physically grounded Bayesian optimization framework for a half-car model. Simulation and optimization tools like these can be leveraged by engineers in the automotive and motorsport sectors to make informed decisions during suspension design. This can save time, money and effort in terms of resources spent on vehicle dynamics development.

It is common practice for motorsport teams participating in championships like Formula 1 to run simulations on their cars on specific race tracks in order to aid race engineers with setup choices. High fidelity simulation and optimization tools are crucial for both off-season and in-season development of a race car and can heavily influence the outcomes of a championship.

10. REFERENCES

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